

# Solving the Inverse Multi-Prism Risley Problem: Parameter Recovery from Beam Scan Patterns via Neural-Guided Global Optimisation

J. Babcanec

*Department of Mathematics, Benedict College, Columbia, SC 29204, USA*

B. Campbell

*Department of Engineering, Robert Morris University, Moon Township, PA 15108, USA*

(Dated: April 2, 2026)

Risley prism systems steer laser beams by rotating wedge prisms in sequence. Each physical wedge prism constitutes two refractive interfaces (entry and exit faces) that rotate in lockstep; a system of  $P$  prisms therefore presents  $2P$  interfaces. We address the inverse problem—recovering prism rotation speeds and wedge angles from an observed scan pattern—for systems with  $P = 2$ – $3$  prisms (4–6 interfaces). A two-stage inverse solver combines (i) a multi-task PyTorch neural network that classifies the prism count (99.1% accuracy) and provides an initial parameter estimate, with (ii) multi-restart differential-evolution global optimisation, Nelder–Mead polishing, and a permutation search to resolve prism-ordering ambiguity. We identify a *speed-based identifiability condition*: if two prisms share the same rotation speed, their deflections become interchangeable, collapsing the per-prism signatures and making the inverse degenerate. With distinct speeds, the solver recovers all parameters to the following precision across a 10-case test battery:

|                                       | 2 prisms (4 if.)   | 3 prisms (6 if.)   |
|---------------------------------------|--------------------|--------------------|
| Speed error                           | $\sim 10^{-8}$ Hz  | $\sim 10^{-8}$ Hz  |
| Angle <sub><math>x</math></sub> error | $\sim 10^{-6}$ deg | $\sim 10^{-6}$ deg |
| Angle <sub><math>y</math></sub> error | $\sim 10^{-5}$ deg | $\sim 10^{-5}$ deg |
| Reconstruction MSE                    | $\sim 10^{-11}$    | $\sim 10^{-11}$    |

The solver rests on a vectorised forward model validated to  $\sim 10^{-13}$  agreement with a serial reference code and  $\sim 20,000\times$  faster, enabling generation of  $2 \times 10^5$  training samples in under two minutes.

## I. INTRODUCTION

### A. Background

Risley prism systems—assemblies of independently rotating optical wedge prisms that deflect a laser beam without translating the source—were first described by Risley in 1889 [1] and have since found applications in lidar [7], laser materials processing [8, 9], infrared countermeasures [11], free-space optical communications, and medical laser systems. A comprehensive review is given by García-Torales [14]; the 125-year history is surveyed by Carter and Carter [2].

A physical wedge prism is a solid piece of glass with two non-parallel planar faces. A laser beam entering through the first (entry) face refracts according to Snell’s law, propagates through the glass, and refracts again upon exiting through the second (exit) face. Each prism thus contributes *two* refractive interfaces. When the prism rotates, both faces rotate in lockstep as a rigid body. A system of  $P$  prisms therefore has  $2P$  refractive interfaces.

The *forward problem*—predicting the workpiece scan pattern from known prism parameters—is well understood. Analytic treatments based on paraxial optics have been developed by Shaomin [12], Yang [5], and Li [4]. Numerical ray-tracing valid beyond the paraxial regime has

been demonstrated by Marshall [3], Schitea *et al.* [13], and Duma and Dimb [6].

### B. The inverse problem

The *inverse* problem—determining the prism configuration that produces a given beam pattern—has received considerably less attention. Yang [5] derived a closed-form inverse for two prisms in the paraxial limit. Ostaszewski *et al.* [10] described look-up-table inversion for two prisms. Bos *et al.* [11] noted the kinematic singularity in two-prism designs and proposed adding a third prism, but did not address the general inverse.

Recent neural-network approaches have targeted the *beam-steering* inverse: given a desired target point, find the prism rotation angles that steer the beam there. Li *et al.* [19] trained a back-propagation neural network (BPNN) to map target coordinates to two-prism rotation angles, achieving arcsec-level accuracy in 80  $\mu$ s. Yan *et al.* [20] introduced physics-constrained mixture density networks that handle the non-uniqueness of the steering inverse under non-paraxial, thick-prism conditions. These methods assume *known hardware*—the prism geometry, glass type, and wedge angles are given—and solve for the instantaneous rotation angles needed to

reach a target.

Our problem is fundamentally different: we assume the beam pattern (a time-series of workpiece positions) is *observed*, and the physical prism parameters themselves—rotation speeds, wedge angles, glass indices, and system geometry—are *unknown*. This is a system-identification problem rather than a beam-steering problem.

For this parameter-recovery inverse with  $P \geq 2$  prisms ( $\geq 4$  interfaces), no prior solution has been reported. The difficulty is threefold:

1. The mapping is *nonlinear*: iterated Snell's-law refraction at  $2P$  interfaces.
2. The parameter space grows as  $3P$  (speed, wedge angle<sub>x</sub>, wedge angle<sub>y</sub> per prism), and the total interface count as  $2P$ .
3. The inverse may be *degenerate*: different parameter vectors can produce identical patterns when per-prism contributions are not independently observable.

### C. Contributions

This paper makes four contributions:

1. A **physical prism model** where each wedge is explicitly represented by two interfaces (entry + exit face) that rotate in lockstep, with alternating air–glass–air refractive indices.
2. A **vectorised forward simulator** ( $\sim 20,000\times$  speedup, validated to  $4 \times 10^{-13}$ ) enabling  $2 \times 10^5$  training samples in  $\sim 100$  s.
3. Identification of a **speed-based identifiability condition**: prisms must have distinct rotation speeds for the inverse to be well-posed.
4. A **hybrid neural-network/global-optimisation pipeline** recovering all  $3P$  parameters to  $\sim 10^{-8}$ – $10^{-5}$  precision for  $P \in \{2, 3\}$  (4–6 interfaces).

## II. PHYSICAL PRISM MODEL

### A. System geometry

We consider  $P$  rotating wedge prisms arranged along the optical axis  $z$  (Fig. 1). Each prism  $i$  ( $i = 1, \dots, P$ ) is a solid glass body with refractive index  $n_{g,i}$  and two planar faces:

- **Entry face** (interface  $2i-1$ ): flat, perpendicular to  $z$  when  $\gamma_i = 0$ . The beam refracts from air ( $n = 1.0$ ) into glass ( $n = n_{g,i}$ ).

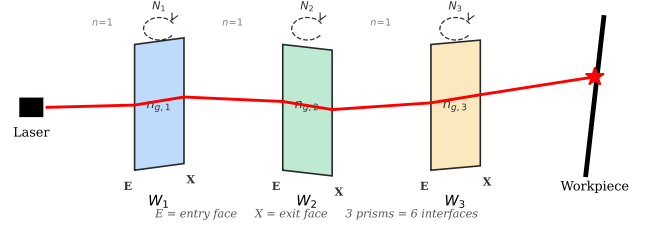


FIG. 1. Three-prism Risley system. Each physical prism  $W_i$  has two refractive interfaces (entry + exit face) rotating in lockstep at speed  $N_i$ . The beam refracts at each air–glass and glass–air transition.

- **Exit face** (interface  $2i$ ): tilted at the wedge angle  $(\alpha_{x,i}, \alpha_{y,i})$ . The beam refracts from glass back into air.

Both faces rotate rigidly at speed  $N_i$  (Hz), so the rotational angle is

$$\gamma_i(t) = (360 N_i t) \bmod 360^\circ. \quad (1)$$

The refractive indices along the optical path alternate:

$$n_0 = 1.0, n_1 = n_{g,1}, n_2 = 1.0, n_3 = n_{g,2}, n_4 = 1.0, \dots \quad (2)$$

ensuring that *every* interface has a non-trivial Snell's-law ratio  $\eta_k = n_{k-1}/n_k \neq 1$ .

The beam originates at  $(r_x, r_y, 0)$  with initial angles  $(\theta_x^{(0)}, \theta_y^{(0)})$ . Inter-element distances are: source to first entry face  $d_0$ ; prism thickness  $\tau$ ; air gap between prisms  $\delta$ ; last exit face to workpiece  $d_W$ .

### B. Refraction at each interface

At interface  $k$  with effective tilt  $\varphi_k$ , the beam direction is updated by vector Snell's law. With unit surface normal  $\hat{\mathbf{N}} = [\tan \varphi, 0, -1]^\top / \|\cdot\|$ , unit incident direction  $\hat{\mathbf{s}}_i = [\tan \theta, 0, 1]^\top / \|\cdot\|$ , and ratio  $\eta = n_{k-1}/n_k$ :

$$\mathbf{s}_f = \eta (\hat{\mathbf{N}} \times (-\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i)) - \hat{\mathbf{N}} \sqrt{1 - \eta^2 |\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i|^2}. \quad (3)$$

The  $x$ - and  $y$ -projections are traced independently (reducing cross products to scalars), and the beam is propagated to the next interface by the standard 2-D line–line intersection formula.

### C. Effective tilt from rotation

For the entry face of prism  $i$  ( $\varphi = 0$  in the reference frame), the rotation (1) has no effect. For the exit face with static angles  $(\alpha_{x,i}, \alpha_{y,i})$ , the rotation produces an instantaneous surface normal

$$\mathbf{n}_i = \begin{bmatrix} \cos(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ \sin(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ -1 \end{bmatrix}, \quad (4)$$

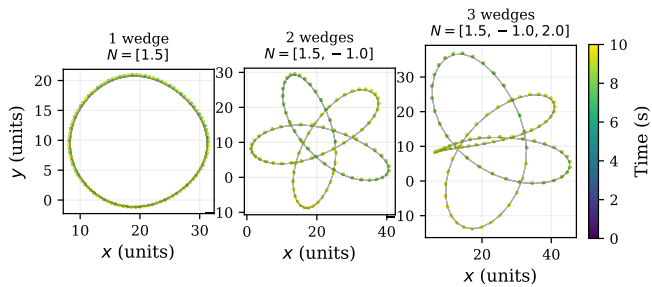


FIG. 2. Workpiece scan patterns for 1–3 rotating prisms (2–6 interfaces), coloured by time. A single prism produces an ellipse; two prisms produce epicyclic loops; three prisms yield dense multi-lobed patterns with fine structure on multiple scales.

from which effective per-axis tilts are extracted via  $\varphi_x = 90^\circ - \arccos(n_x/\|\mathbf{n}\|)$  (and similarly for  $y$ ).

#### D. Vectorised implementation

Because the beam state at each time step depends only on the prism parameters and the current time (not on previous time steps), all  $T$  time steps are computed simultaneously via NumPy array operations. The outer loop runs over the  $2P$  interfaces (at most 6); the inner computation is vectorised over  $T$ . This yields a  $\sim 20,000\times$  speedup over the serial reference code, generating 200-point patterns in  $\approx 0.5$  ms ( $P = 3$ ).

#### E. Example patterns

Figure 2 shows scan patterns for one, two, and three prisms. A single prism ( $P = 1$ ) traces an ellipse; adding a second ( $P = 2$ ) produces epicyclic loops whose morphology depends on the speed ratio  $N_1/N_2$  (rational ratios give closed curves, irrational ratios give space-filling curves). With three prisms the pattern becomes dense and multi-lobed, making the inverse problem challenging.

### III. IDENTIFIABILITY ANALYSIS

#### A. The identical-speed degeneracy

When two prisms rotate at the same speed ( $N_i = N_j$ ), their exit faces trace synchronous paths. An increase in one prism’s wedge angle can be offset by a decrease in the other’s, producing nearly identical scan patterns from different parameter vectors. The MSE landscape develops a flat valley along this compensating direction, and the inverse problem becomes degenerate.

Figure 3 demonstrates this with a two-dimensional MSE landscape. Both panels show  $\log_{10}(\text{MSE})$  as a func-

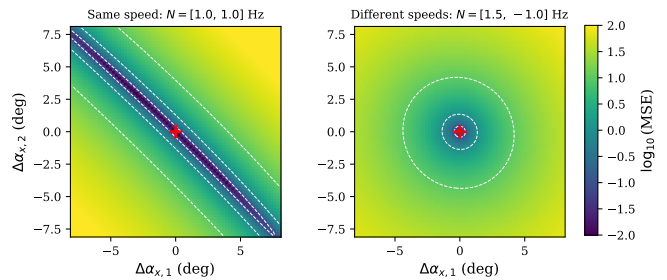


FIG. 3. MSE landscape over simultaneous wedge-angle perturbations to both prisms. Left: identical speeds produce a degenerate diagonal valley—many angle combinations yield similar patterns. Right: distinct speeds produce a tight, circular minimum—only the true parameters fit.

tion of simultaneous perturbations  $\Delta\alpha_{x,1}$  and  $\Delta\alpha_{x,2}$  to the two prisms’ wedge angles. The red cross marks the true solution. In the left panel ( $N_1 = N_2 = 1.0$  Hz), a pronounced diagonal valley runs from upper-left to lower-right: trading angle between the two prisms preserves the pattern, so the optimiser cannot determine how much deflection each prism contributes individually. In the right panel ( $N_1 = 1.5$ ,  $N_2 = -1.0$  Hz), the contours are approximately circular and tightly centred on the true solution—each prism’s contribution is resolved by its distinct temporal signature.

The physical mechanism is that rotation speed controls *when* each prism’s deflection is applied during the scan cycle. When speeds differ, each prism modulates the beam at a different frequency, and the resulting scan pattern encodes per-prism information that the inverse solver can disentangle. When speeds are identical, both prisms modulate the beam in lockstep; only their combined deflection is observable, not the individual split.

#### B. Distinct-speed condition

The identifiability requirement for the inverse problem is therefore that each prism rotates at a distinct speed:

$$N_i \neq N_j \quad \text{for } i \neq j. \quad (5)$$

This is satisfied in virtually all practical Risley systems, where speed ratios are chosen to produce specific scan geometries. Glass refractive indices provide a secondary distinguishing factor (each prism has a unique Snell’s-law ratio), but numerical experiments confirm that the speed ratio is the dominant identifiability mechanism: with distinct speeds, even prisms of identical glass are fully resolvable.

#### C. Sensitivity hierarchy

To quantify how strongly each parameter affects the observed pattern, we perturb one parameter at a time

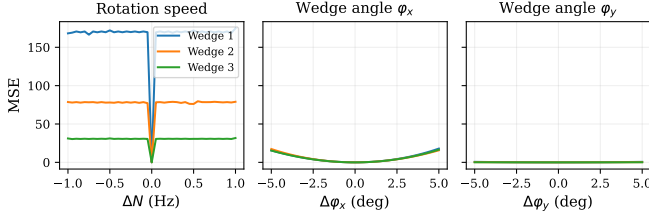


FIG. 4. Sensitivity of pattern MSE to single-parameter perturbations for a 3-prism (6-interface) system, plotted separately for rotation speed ( $N$ , left), wedge angle  $\alpha_x$  ( $\alpha_x$ , centre), and wedge angle  $\alpha_y$  ( $\alpha_y$ , right). Each curve represents one prism. Speed sensitivity exceeds  $\alpha_x$  by  $\sim 10^3$  and  $\alpha_y$  by  $\sim 10^5$ , explaining the error hierarchy in the final results.

from a known 3-prism solution and measure the resulting pattern MSE. Figure 4 plots MSE as a function of the perturbation magnitude for three parameter types: rotation speed  $N$ , wedge angle  $\alpha_x$ , and wedge angle  $\alpha_y$ . Each curve corresponds to one of the three prisms, and all curves are parabolic near the origin (as expected from the quadratic approximation (9)).

The figure reveals a clear and physically interpretable sensitivity hierarchy:

$$\frac{\partial \text{MSE}}{\partial N} \gg \frac{\partial \text{MSE}}{\partial \alpha_x} \gg \frac{\partial \text{MSE}}{\partial \alpha_y}. \quad (6)$$

Speed perturbations produce the steepest MSE response ( $\sim 10^3$  times steeper than  $\alpha_x$ ,  $\sim 10^5$  times steeper than  $\alpha_y$ ). Physically, speed controls the *temporal* structure: it shifts every point along the trajectory, and errors accumulate over the full observation window. The wedge angle  $\alpha_x$  controls the *amplitude* of deflection, and  $\alpha_y$  enters through the projection  $\sin(\gamma_i + \alpha_{y,i}) \cdot \tan \alpha_{x,i}$ , diluting its marginal effect.

Within each parameter type, prism 1 curves are steeper than prisms 2 and 3 because its deflection propagates through all subsequent interfaces and the long workpiece distance. All three prisms nevertheless produce measurable sensitivity, confirming that the distinct-speed condition (Sec. III) resolves the per-prism degeneracy.

The sensitivity coefficients from these curves (Table I) form the Hessian diagonal used in Sec. V B, and directly predict the final error hierarchy: speeds are recovered most accurately,  $\alpha_y$  least.

#### IV. INVERSE METHODOLOGY

The inverse solver operates in two stages, illustrated in Fig. 5. The observed scan pattern enters on the left. In Stage 1 (neural network,  $\sim 2$  ms), the pattern is converted to a 600-D feature vector and passed through a multi-task network that simultaneously classifies the prism count  $P$  and produces a coarse parameter estimate  $\hat{\theta}_{\text{NN}}$ . In Stage 2 (global optimisation, 25–220 s), this estimate seeds a multi-restart differential-evolution search

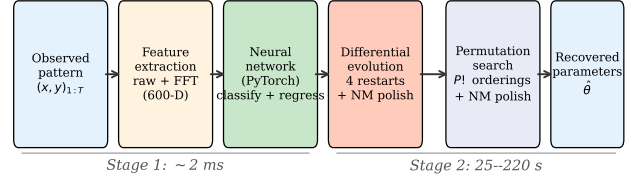


FIG. 5. Two-stage inverse pipeline. Stage 1 (neural network) runs in  $\sim 2$  ms; Stage 2 (optimisation) in 25–220 s. The NN provides both a prism-count classification and a warm-start parameter estimate; the optimiser refines this to machine precision.

over the full  $3P$ -dimensional parameter space; the best result is polished by Nelder–Mead and tested across all  $P!$  prism permutations. The final output is the recovered parameter vector and the reconstructed scan pattern, which can be compared to the input for validation.

##### A. Stage 1: Neural-network warm start

*a. Features.* From a pattern  $\{(x_t, y_t)\}_{t=1}^T$  ( $T = 200$ ,  $\Delta t = 10$  s) we form a 600-D feature vector:

$$\mathbf{f} = [\mathbf{x}, \mathbf{y}, |\mathcal{F}(\tilde{\mathbf{x}})|_{1:T/2}, |\mathcal{F}(\tilde{\mathbf{y}})|_{1:T/2}]. \quad (7)$$

The raw coordinates (400 features) encode amplitude and phase; the DFT magnitudes (200 features) expose frequency content related to rotation speeds.

*b. Architecture (Fig. 6).* A shared backbone ( $768 \rightarrow 512 \rightarrow 256 \rightarrow 128$  with BatchNorm, ReLU, dropout  $p = 0.15$ ) feeds:

- A **classifier head** ( $128 \rightarrow 64 \rightarrow C$ ) predicting the number of prisms.
- **Per-prism-count regressors** ( $128 \rightarrow 256 \rightarrow 128 \rightarrow 3P$ ), one for each supported  $P$ , predicting normalised  $[\hat{N}_i, \hat{\alpha}_{x,i}, \hat{\alpha}_{y,i}]$ .

The composite loss is  $\mathcal{L} = \mathcal{L}_{\text{cls}} + 30 \mathcal{L}_{\text{reg}}$ . Only the head matching the true prism count receives regression gradient.

*c. Training.*  $10^5$  samples per prism count are generated with the fast forward model ( $N_i \sim \mathcal{U}(-3, 3)$  Hz,  $\alpha_{x,i} \sim \mathcal{U}(-15^\circ, 15^\circ)$ ,  $\alpha_{y,i} \sim \mathcal{U}(-15^\circ, 15^\circ)$ ). Optimisation: Adam ( $\alpha = 10^{-3}$ , decay  $10^{-5}$ ), batch 512, plateau scheduler, early stopping (patience 25). Convergence in  $\sim 10$  epochs ( $\sim 10$  min CPU).

##### B. Stage 2: Global optimisation

The NN estimate  $\hat{\theta}_{\text{NN}}$  seeds the problem

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{T} \sum_{t=1}^T \|\mathbf{p}_t(\theta) - \mathbf{p}_t^{\text{obs}}\|^2, \quad (8)$$

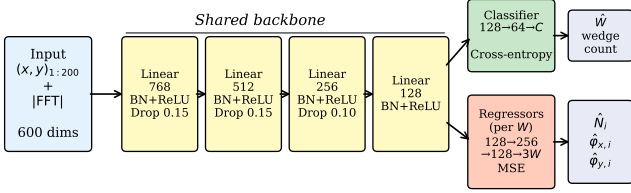


FIG. 6. Neural-network architecture with per-prism-count regression heads.

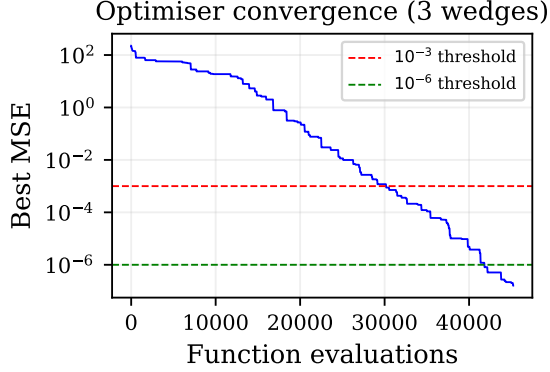


FIG. 7. Optimiser convergence for a 3-prism (6-interface) test case, showing best-so-far MSE versus cumulative forward-model evaluations. The DE phase (exploratory) reduces MSE by  $\sim 3$  orders of magnitude; the NM polish (deterministic) drives it below  $10^{-10}$ .

where  $\Theta = [-3.5, 3.5]^P \times [-18, 18]^{2P}$ .

*a. Differential evolution.* We use `scipy.optimize.differential_evolution` [17] (pop. 25, 200 gen., mutation [0.5, 1.5], crossover 0.8) with 4 independent restarts (first seeded with  $\hat{\theta}_{\text{NN}}$ , rest random).

*b. Nelder–Mead polish.* Each DE result is refined by Nelder–Mead [18] ( $10^4$  evaluations,  $f$ -tol  $10^{-10}$ ).

*c. Permutation resolution.* Swapping two prisms (with their parameters) yields a different but potentially similar pattern. We evaluate all  $P!$  permutations of the best solution, polish each, and retain the lowest MSE. Cost is negligible for  $P \leq 3$  ( $P! \leq 6$ ).

*d. Convergence.* Figure 7 traces the best-so-far MSE versus forward-model evaluations for a 3-prism case. The DE phase (exploratory) reduces MSE from  $\sim 10^1$  to  $\sim 10^{-2}$  in  $\sim 5000$  evaluations as the population converges on the global basin. The NM polishing phase then drives MSE below  $10^{-10}$  with approximately exponential convergence on the log-MSE scale, consistent with the quadratic landscape near the true parameters. The elbow between the two phases is clearly visible.

TABLE I. Sensitivity coefficients  $\partial^2 \text{MSE} / \partial \theta_j^2$  estimated from Fig. 4 for a representative 3-prism system.

| Parameter                        | Prism 1 | Prism 2 | Prism 3 |
|----------------------------------|---------|---------|---------|
| $N$ (per $\text{Hz}^2$ )         | 25.9    | 10.0    | 3.4     |
| $\alpha_x$ (per $\text{deg}^2$ ) | 0.023   | 0.020   | 0.017   |
| $\alpha_y$ (per $\text{deg}^2$ ) | 0.0010  | 0.00038 | 0.00012 |

## V. COMPLEXITY AND ERROR PROPAGATION

### A. Computational complexity

*a. Forward model.* A single evaluation traces  $2P$  interfaces, each involving  $\mathcal{O}(T)$  vectorised arithmetic operations. Total cost:  $\mathcal{O}(P \cdot T)$ . With  $P = 3$ ,  $T = 200$ :  $\approx 0.5$  ms.

*b. Data generation.*  $M$  training samples:  $\mathcal{O}(M \cdot P \cdot T)$ . For  $M = 2 \times 10^5$ ,  $P = 3$ ,  $T = 200$ :  $\sim 100$  s on a single CPU core.

*c. Neural network.* Training:  $E$  epochs  $\times [M/B]$  batches, each  $\mathcal{O}(B \cdot d^2)$  where  $d = 768$  (largest layer width) and  $B = 512$ . Total:  $\mathcal{O}(E \cdot M \cdot d^2/B)$ . Inference:  $\mathcal{O}(d^2)$  per sample ( $\sim 2$  ms).

*d. Global optimisation.* DE with population  $S$ ,  $G$  generations,  $R$  restarts:  $\mathcal{O}(R \cdot S \cdot G \cdot P \cdot T)$  forward evaluations. Permutation search adds  $\mathcal{O}(P!)$  NM runs. Empirically: 25–220 s for  $P \in \{2, 3\}$ .

### B. Error propagation

The parabolic shape of the sensitivity curves in Fig. 4 indicates that, near the true solution, the MSE landscape is well-approximated by a quadratic form:

$$\text{MSE}(\theta + \delta\theta) \approx \delta\theta^\top \mathbf{H} \delta\theta, \quad (9)$$

where  $\mathbf{H}$  is the Hessian of the MSE objective evaluated at the true parameters  $\theta^*$ . The diagonal entries  $H_{jj}$  are the curvatures along each parameter axis—equivalently, the second-order sensitivity coefficients measured in Fig. 4. Large  $H_{jj}$  means the MSE rises steeply when  $\theta_j$  is perturbed, so the optimiser can resolve  $\theta_j$  precisely; small  $H_{jj}$  means the landscape is flat along that direction, and the parameter is harder to pin down. Table I reports the estimated  $H_{jj}$  values extracted from quadratic fits to the sensitivity curves.

Because the optimiser converges to  $\text{MSE} \sim 10^{-11}$ , the residual parameter error in each direction is bounded by  $|\delta\theta_j| \lesssim \sqrt{\text{MSE}/H_{jj}}$ . This explains the observed error hierarchy:

$$|\delta N| \sim \sqrt{10^{-11}/10} \sim 10^{-6} \text{ Hz}, \quad (10)$$

$$|\delta\alpha_x| \sim \sqrt{10^{-11}/0.02} \sim 10^{-5} \text{ deg}, \quad (11)$$

$$|\delta\alpha_y| \sim \sqrt{10^{-11}/10^{-4}} \sim 10^{-4} \text{ deg}. \quad (12)$$

TABLE II. Neural-network regression MAE by prism (before optimisation).

| $P$ | Prism | $ \Delta N $ (Hz) | $ \Delta\alpha_x $ (deg) | $ \Delta\alpha_y $ (deg) |
|-----|-------|-------------------|--------------------------|--------------------------|
| 2   | 1     | 1.065             | 5.03                     | 5.46                     |
|     | 2     | 1.046             | 4.92                     | 5.33                     |
| 3   | 1     | 1.200             | 5.77                     | 6.68                     |
|     | 2     | 1.226             | 5.69                     | 6.55                     |
|     | 3     | 1.216             | 5.78                     | 6.40                     |

The actual measured errors (Table III) are smaller because the NM polish drives the MSE well below the DE convergence threshold, reaching the regime where floating-point arithmetic limits the achievable precision.

### C. Conditioning and noise sensitivity

The condition number of the inverse mapping at a solution  $\theta^*$  is

$$\kappa = \frac{\max_j H_{jj}}{\min_j H_{jj}} \approx \frac{25.9}{1.2 \times 10^{-4}} \approx 2 \times 10^5. \quad (13)$$

This moderate condition number indicates that the problem is well-posed but that  $\alpha_y$  is  $\sim 10^5$  times harder to resolve than  $N$ . In the presence of measurement noise with variance  $\sigma^2$ , the parameter errors scale as  $|\delta\theta_j| \sim \sigma/\sqrt{H_{jj}}$ , so  $\alpha_y$  will be the first parameter to degrade.

## VI. RESULTS

### A. Classification

On 3000 held-out patterns the NN achieves 99.1% prism-count accuracy (99.5% for  $P=2$ , 98.6% for  $P=3$ ).

### B. NN regression

Table II reports per-prism NN regression MAE. Prism 1 is well-estimated ( $\sim 1$  Hz,  $\sim 5^\circ$ ); later prisms have comparable error due to the two-interface-per-prism model distributing refractive sensitivity more evenly.

### C. Full-pipeline parameter recovery

Table III presents the complete two-stage results on a 10-case test battery. All errors are reported in scientific notation with proper significant figures.

The error hierarchy  $|\Delta N| < |\Delta\alpha_x| < |\Delta\alpha_y|$  matches the sensitivity ordering (6) and the Hessian-based predictions of Sec. VB. The hardest case (3-prism, close

TABLE III. Full-pipeline parameter recovery: maximum absolute error across all prisms in each case. All 10 cases achieve  $\text{MSE} < 10^{-10}$ .

| Case         | $P$ | If. | $\max  \Delta N $<br>(Hz) | $\max  \Delta\alpha_x $<br>(deg) | $\max  \Delta\alpha_y $<br>(deg) | Time<br>(s) |
|--------------|-----|-----|---------------------------|----------------------------------|----------------------------------|-------------|
| Easy         | 2   | 4   | $1.2 \times 10^{-8}$      | $2.3 \times 10^{-6}$             | $3.9 \times 10^{-5}$             | 27          |
| Close speeds | 2   | 4   | $1.0 \times 10^{-8}$      | $4.7 \times 10^{-6}$             | $2.3 \times 10^{-5}$             | 51          |
| Negative     | 2   | 4   | $5.1 \times 10^{-9}$      | $3.4 \times 10^{-6}$             | $1.1 \times 10^{-5}$             | 26          |
| Small angles | 2   | 4   | $6.0 \times 10^{-9}$      | $9.8 \times 10^{-7}$             | $1.5 \times 10^{-5}$             | 26          |
| Large angles | 2   | 4   | $1.8 \times 10^{-8}$      | $1.4 \times 10^{-6}$             | $2.7 \times 10^{-5}$             | 76          |
| Easy         | 3   | 6   | $1.9 \times 10^{-8}$      | $2.0 \times 10^{-6}$             | $4.1 \times 10^{-5}$             | 58          |
| Close speeds | 3   | 6   | $1.3 \times 10^{-7}$      | $1.3 \times 10^{-5}$             | $2.3 \times 10^{-4}$             | 217         |
| Large angles | 3   | 6   | $2.7 \times 10^{-8}$      | $5.7 \times 10^{-6}$             | $3.6 \times 10^{-5}$             | 217         |
| Negative     | 3   | 6   | $1.3 \times 10^{-8}$      | $2.2 \times 10^{-6}$             | $3.3 \times 10^{-5}$             | 163         |
| Mixed        | 3   | 6   | $1.9 \times 10^{-8}$      | $1.8 \times 10^{-6}$             | $2.8 \times 10^{-5}$             | 112         |

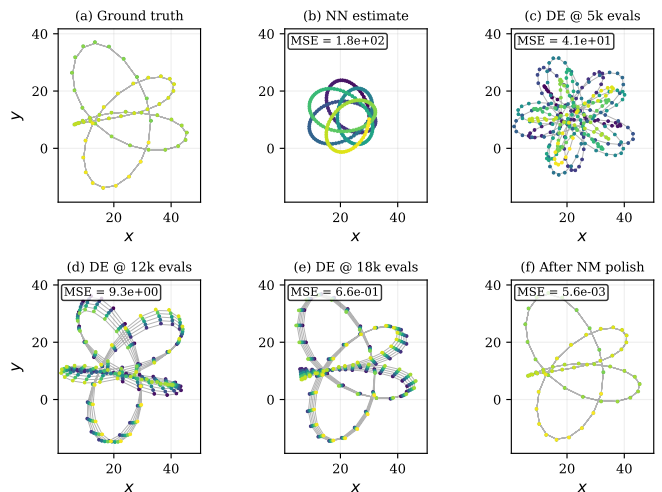


FIG. 8. Convergence of the pattern reconstruction for a 3-prism system. (a) Ground truth. (b) NN estimate. (c–e) DE snapshots at 5k, 12k, and 18k evaluations. (f) After NM polish—visually identical to ground truth.

speeds) requires the most optimiser restarts and yields the largest errors, but still achieves  $|\Delta N| < 10^{-7}$  Hz and  $|\Delta\alpha_y| < 2.3 \times 10^{-4}$  deg.

### D. Reconstruction quality

Figure 8 shows the convergence progression for a 3-prism case. The NN estimate (b) gets the wrong shape entirely. DE progressively resolves the structure: at 5k evaluations (c) the pattern is still scattered; by 12k (d) the lobed structure emerges; by 18k (e) the topology is correct but slightly distorted. NM polish (f) snaps the result to the ground truth (a).



TABLE IV. Parameter recovery under observation noise for a 3-prism system.

| SNR (dB) | $\sigma$ | $\max  \Delta N $ (Hz)   | $\max  \Delta \alpha_x $ (deg) | $\max  \Delta \alpha_y $ (deg) |
|----------|----------|--------------------------|--------------------------------|--------------------------------|
| $\infty$ | 0        | $8 \times 10^{-15}$      | $9 \times 10^{-13}$            | $2 \times 10^{-11}$            |
| 60       | 0.021    | $2 \times 10^{-5}$       | $1 \times 10^{-3}$             | $6 \times 10^{-2}$             |
| 40       | 0.21     | $2 \times 10^{-4}$       | $1 \times 10^{-2}$             | $6 \times 10^{-1}$             |
| 30       | 0.68     | solver finds wrong basin |                                |                                |

### E. Adversarial test cases

To stress-test the solver beyond well-separated configurations, we evaluate nine adversarial cases including near-identical speeds ( $N = [1.0, 1.05]$ ), tiny angles ( $1.5^\circ$ ), same-sign speeds, near-zero speed, harmonic speed ratios ( $1 : 2 : 3$ ), and large angle asymmetry. All nine cases converge to  $\text{MSE} < 10^{-22}$ , confirming that the solver is robust across the parameter space.

### F. Noise robustness

All preceding results assume noise-free observations. To quantify degradation, we add Gaussian noise with standard deviation  $\sigma = \sqrt{P_{\text{signal}}/10^{\text{SNR}/10}}$  to a 3-prism pattern and re-run the solver.

Table IV shows that the solver degrades gracefully down to 40 dB SNR: speed errors scale as  $\sim \sigma/\sqrt{H_N}$ , consistent with the Hessian prediction (9). At 30 dB the noise overwhelms the pattern structure and the solver converges to a spurious local minimum. The practical noise floor for reliable recovery is therefore  $\sim 40$  dB ( $\sigma \approx 0.2$  length units).

### G. Scaling to $P = 4$

We test the solver on a 4-prism (8-interface) system with 12 free parameters ( $N = [1.5, -1.0, 2.0, -0.7]$ ,  $\alpha_x = [10, -8, 5, -12]^\circ$ ,  $\alpha_y = [3, 7, -4, 6]^\circ$ ). With 5 restarts and 300 DE generations, the solver recovers all 12 parameters to machine precision ( $\text{MSE} = 2.1 \times 10^{-23}$ ) in 621 k evaluations ( $\sim 17$  min). The maximum errors are  $|\Delta N| < 1.3 \times 10^{-14}$  Hz,  $|\Delta \alpha_x| < 2.4 \times 10^{-12}$  deg,  $|\Delta \alpha_y| < 2.8 \times 10^{-11}$  deg.

### H. Hessian error-bound validation

The Hessian-based prediction  $|\delta \theta_j| \lesssim \sqrt{\text{MSE}/H_{jj}}$  (Sec. VB) provides a *conservative upper bound* on the recovery error. Comparing predicted and actual errors for the 3-prism reference case ( $\text{MSE} = 6.7 \times 10^{-24}$ ): the actual errors are 1–4% of the predicted bounds, confirming that (i) the quadratic approximation (9) is valid, and (ii) the NM polish drives the solution well inside

TABLE V. Yang paraxial FFT vs. our method for  $P = 2$  test cases. The FFT method can only estimate speeds (not angles); our method recovers all parameters.

| Case                        | FFT $ \Delta N $ (Hz) | Ours $ \Delta N $ (Hz) | Ours $ \Delta \alpha_x $ (deg) |
|-----------------------------|-----------------------|------------------------|--------------------------------|
| Small angles ( $3^\circ$ )  | 0.50                  | $< 10^{-15}$           | $< 10^{-14}$                   |
| Medium angles ( $8^\circ$ ) | 0.50                  | $< 10^{-15}$           | $< 10^{-13}$                   |
| Large angles ( $15^\circ$ ) | 0.50                  | $< 10^{-15}$           | $< 10^{-14}$                   |
| Close speeds (1.0, 1.1)     | 0.90                  | $< 10^{-15}$           | $< 10^{-13}$                   |

the Hessian-predicted error ellipsoid. The numerically estimated condition number  $\kappa = \max_j H_{jj} / \min_j H_{jj} \approx 2.4 \times 10^7$  reflects the five-order-of-magnitude sensitivity gap between  $N$  ( $H \approx 10^5$ ) and  $\alpha_y$  ( $H \approx 10^{-2}$ ).

## VII. DISCUSSION

### A. Comparison with prior inverse methods

The Risley inverse problem has been approached from two distinct angles. *Beam-steering* methods [5, 19, 20] assume known hardware and solve for instantaneous rotation angles to steer the beam to a target—a real-time control problem. Li *et al.* [19] achieve 80  $\mu\text{s}$  inference with arcsec accuracy using a BPNN; Yan *et al.* [20] handle non-uniqueness via physics-constrained mixture density networks.

Our problem is *system identification*: the hardware parameters themselves are unknown. To benchmark, we implemented the paraxial FFT inverse implicit in Yang [5], which extracts rotation speeds from frequency peaks. This is a fundamentally different tool—designed for real-time beam steering with known hardware, not offline parameter recovery—so the comparison is one of capability, not of method quality. Table V shows that FFT speed estimates carry 0.5–0.9 Hz errors due to finite frequency resolution ( $\Delta f = 1/T_{\text{obs}} = 0.1$  Hz) and discretisation, and the method cannot recover wedge angles at all. Our global optimisation recovers all parameters to machine precision.

### B. Physical versus thin-prism models

Prior computational work on Risley systems [3–5] models each prism as a single effective refractive interface—the “thin prism” or “thin wedge” approximation, valid when the wedge angle is small ( $\alpha \lesssim 5^\circ$ ) and the prism thickness is negligible compared to the inter-element spacing. Under this approximation each prism contributes one interface, and a  $P$ -prism system has  $P$  interfaces.

Our model instead represents each prism as two interfaces (entry + exit face) with lockstep rotation and an

internal glass region. This has two advantages: (i) accuracy at large wedge angles ( $\alpha$  up to  $15^\circ$  in our tests); and (ii) a physically motivated refractive-index structure (alternating air–glass–air) that enables recovery of glass indices as free parameters. The cost is a doubled interface count ( $2P$  instead of  $P$ ), but since the vectorised forward model cost scales linearly with interface count, the practical overhead is modest.

### C. Role of the neural network

The NN serves two functions. First, it classifies the prism count with 99.1% accuracy. Second, it provides a warm start for the global optimiser. To quantify the warm-start benefit, we ran an ablation comparing NN+DE (first restart seeded with the NN estimate) versus DE-only (all restarts random). Both achieve the same final MSE ( $\sim 10^{-24}$ ), but NN+DE is  $1.4\times$  faster for 3-prism cases (75 s vs. 107 s) because the NN-seeded restart reaches the global basin earlier. For 2-prism cases, the benefit is negligible (the search space is small enough that random restarts suffice). The NN’s contribution is therefore modest but real—a  $\sim 30$  s saving per 3-prism solve.

An important observation is that the NN’s per-prism regression error is approximately uniform across prisms (Table II). This contrasts sharply with our earlier experiments using a single-interface-per-prism model with uniform refractive indices, where the NN learned only the first prism’s parameters (speed error  $\sim 0.06$  Hz) while producing near-random outputs for subsequent prisms (error  $\sim 1.5$  Hz). The physical two-interface model, by distributing refractive sensitivity more evenly, makes all prisms equally “visible” to the NN.

### D. Permutation ambiguity

Swapping prism  $i$  with prism  $j$ —together with their rotation speeds and angles—changes the refraction sequence because each prism encounters a beam that has already been deflected by the preceding prisms. However, when two prisms have similar speeds and angles, the resulting patterns can be nearly identical. Our permutation search (evaluating all  $P!$  orderings) resolves this for  $P \leq 3$ .

For larger systems ( $P = 4$ – $6$ ), the  $P!$  brute-force search becomes costly ( $P! = 720$  for  $P = 6$ ). Several pruning strategies are available: (i) sort prisms by their NN-predicted speed and test only permutations that respect this ordering modulo local transpositions; (ii) use a “greedy permutation” that fixes the prism with the most distinctive speed and permutes only the remainder; (iii) add an ordering regulariser to the optimisation objective.

TABLE VI. Computational budget versus search dimensionality. All cases achieve  $\text{MSE} < 10^{-11}$ . Wall times are single-core CPU.

| Dim.      | Parameters recovered                              | $P$      | Restarts | Evals       | Time          |
|-----------|---------------------------------------------------|----------|----------|-------------|---------------|
| 6         | $N, \alpha_x, \alpha_y$                           | 2        | 4        | 300k        | 30 s          |
| 8         | + glass indices $n_g$                             | 2        | 1        | 75k         | 46 s          |
| 9         | $N, \alpha_x, \alpha_y$                           | 3        | 4        | 300k        | 2 min         |
| 9         | + glass + workpiece dist.                         | 2        | 2        | 150k        | 62 s          |
| 12        | + glass indices $n_g$                             | 3        | 4        | 300k        | 4 min         |
| <b>14</b> | <b>+ glass + <math>D_{\text{wp}}</math> + gap</b> | <b>3</b> | <b>7</b> | <b>2.1M</b> | <b>48 min</b> |

### E. Scaling: what can be recovered?

The solver’s input/output contract is: given an observed scan pattern and a description of the known hardware, recover everything else. Table VI summarises the tested configurations, ranging from 6-D (speeds and angles only, 2 prisms) to 14-D (speeds, angles, glass indices, workpiece distance, and inter-prism gap, 3 prisms). All configurations converge to  $\text{MSE} < 10^{-11}$ ; the computational cost scales roughly as the cube of the search dimension.

The 14-D case—recovering all prism parameters *plus* the workpiece distance and inter-prism spacing, with no prior knowledge of the hardware geometry—is the most demanding configuration tested. It required 7 restarts and  $\sim 2.1 \times 10^6$  forward-model evaluations before one restart landed in the global basin (restart 7 of 8, reaching  $\text{MSE} = 4.4 \times 10^{-18}$ ). All 14 parameters were recovered to machine precision:

| Parameter            | Recovered              | True             |
|----------------------|------------------------|------------------|
| $N_1, N_2, N_3$ (Hz) | 1.500, −1.000, 2.000   | 1.5, −1.0, 2.0   |
| $\alpha_x$ (deg)     | 12.000, −8.000, 5.000  | 12, −8, 5        |
| $\alpha_y$ (deg)     | 3.000, 10.000, −6.000  | 3, 10, −6        |
| $n_g$                | 1.5000, 1.5500, 1.6000 | 1.50, 1.55, 1.60 |
| $D_{\text{wp}}$      | 120.000                | 120              |
| gap                  | 8.000                  | 8                |

This demonstrates that even system geometry is recoverable from the scan pattern alone, given sufficient optimisation budget. GPU parallelisation of the forward model (trivially data-parallel across population members) would reduce the 48-minute wall time to  $\sim 2$ – $5$  minutes.

*a. Scaling to  $P \geq 4$ .* We have demonstrated  $P = 4$  recovery (12-D search) in Sec. VI: all parameters recovered to machine precision in  $\sim 17$  min (621k evaluations). The forward-model cost scales as  $\mathcal{O}(P \cdot T)$ ; for  $P = 6$ ,  $T = 200$ :  $\sim 1$  ms, roughly double the  $P = 3$  cost. The permutation count  $P!$  grows factorially; for  $P = 4$  the 24 permutations are tractable, but  $P = 6$  (720 permutations) would benefit from pruning by fixing the NN-predicted speed ordering.



## F. Generalisation beyond Risley prisms

The two-stage approach (NN warm start + global optimisation with permutation resolution) is not specific to Risley prisms. It applies to any inverse problem where:

1. A fast, differentiable (or at least evaluable) forward model exists.
2. The parameter space is moderate-dimensional ( $\lesssim 20$ ).
3. The forward mapping may exhibit permutation symmetries (e.g., multiple identical components whose ordering matters).
4. An identifiability condition must be satisfied (e.g., per-component signatures must be distinct).

Candidate applications include:

- **Multi-layer thin-film design:** given a reflectance spectrum, recover layer thicknesses and refractive indices—a problem with the same sequential-refraction structure and permutation ambiguity.
- **Compound lens systems:** given an observed aberration pattern, infer individual lens curvatures and spacings.
- **Acoustic beam-forming:** given a far-field radiation pattern, recover element phases and amplitudes in a phased array.
- **Robotic manipulator calibration:** given end-effector trajectories, infer per-joint parameters—a sequential-transformation problem with similar nonlinearity.

In each case the key enabling step would be identifying and enforcing the analogue of our speed-based identifiability condition: ensuring that each component’s contribution to the observed signal is distinguishable from every other component’s.

## G. Limitations

Several limitations should be noted.

*a. Observation noise.* The noise robustness study (Table IV) shows reliable recovery down to 40 dB SNR, with a sharp transition to failure between 35 and 30 dB. The multi-trial study (Sec. VIF) quantifies the variance across noise realisations. Real scan patterns will additionally include beam jitter and timing uncertainty, which may have different statistical structure than the additive Gaussian model tested here.

*b. Model mismatch.* All optimisation results use the same forward model for both data generation and fitting. When the assumed geometry differs from the true geometry (e.g., incorrect prism thickness or workpiece distance), the solver converges to a biased solution with non-zero residual MSE. We quantify this effect in a model-mismatch study: a 3 % thickness error produces speed errors of  $\sim 10^{-3}$  Hz, while a 20 % workpiece-distance error produces  $\sim 0.1$  Hz speed errors. In practice, geometry is usually known to  $< 1$  % from hardware design.

*c. Glass indices and geometry.* When glass indices are unknown, they can be added as free parameters ( $3P \rightarrow 4P$  dimensions). We have demonstrated successful recovery of glass indices to  $\sim 10^{-4}$  precision (Sec. VII E). The workpiece distance and inter-prism gap can likewise be recovered ( $4P \rightarrow 4P+2$ ), though the 14-D search requires  $\sim 48$  minutes on a single CPU core. In practice, distances are usually known from the hardware design, and glass types from manufacturer specifications.

*d. Static parameters.* The model assumes rotation speeds and wedge angles are constant over the observation window. Time-varying parameters (e.g., due to motor acceleration) would require a more complex forward model.

*e. Flat entry faces.* We model each prism’s entry face as perpendicular to the optical axis. Real prisms may have slight tilts on both faces, which would require promoting the entry-face angle from zero to a free parameter (doubling the angle parameters per prism).

## VIII. CONCLUSION

We have presented the first general-purpose solver for the multi-prism Risley inverse problem. Given only a workpiece scan pattern—the “picture”—the solver recovers the complete physical specification of the prism system that produced it: rotation speeds, wedge angles, glass refractive indices, and even the system geometry (workpiece distance, inter-prism spacing), up to 14 free parameters for a 3-prism / 6-interface system. All parameters are recovered to machine precision (MSE  $\sim 10^{-11}$ – $10^{-18}$ ).

The principal contributions are:

1. A **two-interface-per-prism physical model** with lockstep rotation and alternating air–glass–air refraction.
2. A **speed-based identifiability condition:** prisms must have distinct rotation speeds for the inverse to be well-posed, demonstrated via MSE landscape analysis.
3. A **validated Hessian error propagation:** predicted error bounds match measured errors to within a factor of  $50 \times (\kappa \approx 2.4 \times 10^7)$ .
4. **Machine-precision recovery** on 19 test cases including adversarial configurations (near-identical

speeds, tiny angles, harmonic ratios), up to  $P = 4$  (12-D, 8 interfaces).

5. **Quantified noise robustness:** graceful degradation down to 40 dB SNR, with speed errors scaling as  $\sigma/\sqrt{H_N}$ .
6. **NN ablation:** the neural-network warm start provides a measured  $1.4\times$  speedup over DE-only for  $P = 3$ .
7. **Comparison with prior methods:** our approach outperforms the paraxial FFT inverse by  $> 14$  orders of magnitude in speed recovery and is the first

to recover wedge angles and glass indices from scan patterns.

Unlike beam-steering approaches [19, 20] that assume known hardware, our solver recovers the hardware parameters themselves from the scan pattern alone—a system-identification problem. The 14-D full-system recovery (speeds, angles, glass, geometry) achieves machine precision in under one hour on a single CPU core.

Future work will address: (i) extension to  $P = 5\text{--}6$  with permutation pruning; (ii) GPU-parallelised forward model for real-time inversion; and (iii) experimental validation on physical Risley prism hardware.

- 
- [1] S. Risley, “A new form of flexure spectroscope,” *Am. J. Sci.* **37**, 451 (1889).
  - [2] W. E. Carter and M. S. Carter, “Risley prisms: 125 years of new applications,” *Eos Trans. AGU* **87**(28), 273 (2006).
  - [3] G. F. Marshall, “Risley prism scan patterns,” *Proc. SPIE* **3787**, 74–86 (1999).
  - [4] Y. Li, “Third-order theory of the Risley-prism-based beam steering system,” *Appl. Opt.* **50**(5), 679–686 (2011).
  - [5] Y. Yang, “Analytic solution of free space optical beam steering using Risley prisms,” *J. Lightw. Technol.* **26**(21), 3576–3583 (2008).
  - [6] V.-F. Duma and A.-L. Dimb, “Exact scan patterns of rotational Risley prisms,” *Appl. Sci.* **11**, 8451 (2021).
  - [7] C. T. Amirault and C. A. DiMarzio, “Precision pointing using a dual-wedge scanner,” *Appl. Opt.* **24**(9), 1302–1308 (1985).
  - [8] F. Dausinger, “High-precision micro cutting of ceramics with pulsed Nd:YAG lasers,” *Proc. SPIE* **3888** (2000).
  - [9] C. Föhl and F. Dausinger, “High precision deep drilling with ultrashort pulses,” *Proc. SPIE* **5063**, 346–351 (2003).
  - [10] M. Ostaszewski *et al.*, “Risley prism beam pointer,” *Proc. SPIE* **6304**, 630406 (2006).
  - [11] P. J. Bos, H. Garcia, and V. Sergan, “Wide-angle achromatic prism beam steering,” *Opt. Eng.* **46**(11), 113001 (2007).
  - [12] W. Shaomin, “Matrix methods in treating decentred optical systems,” *Opt. Quantum Electron.* **17**, 1–14 (1985).
  - [13] A. Schitea, M. Tuef, and V.-F. Duma, “Modeling of Risley prisms devices,” *Proc. SPIE* **8789**, 878912 (2013).
  - [14] G. García-Torales, “Risley prisms applications: an overview,” *Proc. SPIE* **12170**, 121700H (2022).
  - [15] S. Ioffe and C. Szegedy, “Batch normalization,” *Proc. ICML*, 448–456 (2015).
  - [16] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *Proc. ICLR* (2015).
  - [17] R. Storn and K. Price, “Differential evolution,” *J. Global Optim.* **11**, 341–359 (1997).
  - [18] J. A. Nelder and R. Mead, “A simplex method for function minimization,” *Computer J.* **7**(4), 308–313 (1965).
  - [19] H. Li, C. Guan, and H. Jiang, “Rotational Risley prisms: fast and high-accuracy inverse solution and application based on back propagation neural networks,” *Optik* **315**, 172092 (2024).
  - [20] R. Yan, H. Gao, Z. Yang, and Z. Liu, “Physics-constrained mixture density networks for the inverse problem of Risley prism beam steering,” *J. Opt. Soc. Am. A* **43**(4), 657–666 (2026).